

Unit-6

Cyber Forensic, Hacking& its countermeasures

- Cyber Forensics: <https://youtu.be/2D5wTo1adbg>
is a field of technology that uses investigation techniques to help identify, collect, and store evidence from an electronic device.
- Hacking:
is the act of compromising digital devices and networks through unauthorized access to an account or computer system.
it is most commonly associated with illegal activity and data theft by cyber criminals.

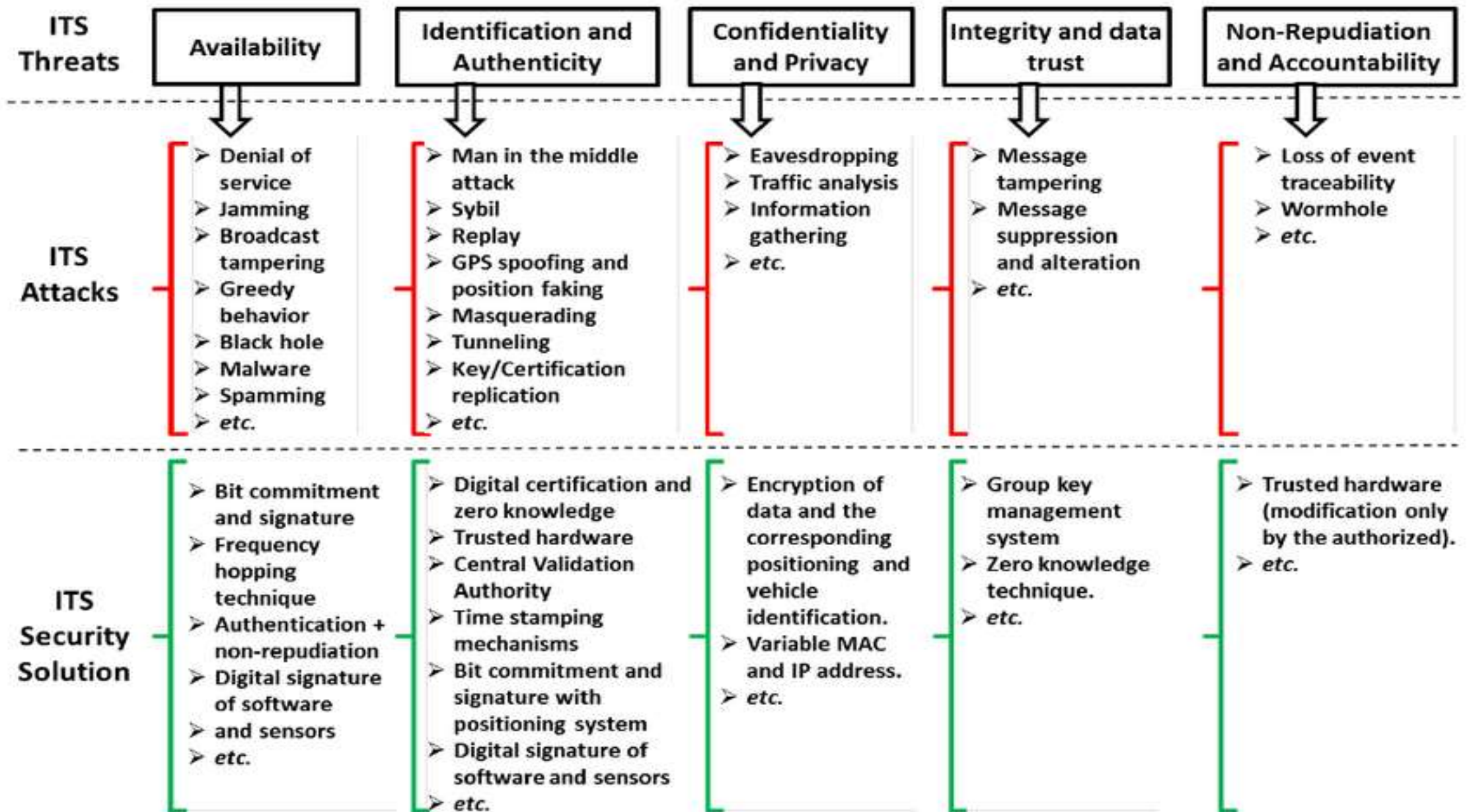
Black Hat Hackers



Motivation: Steal personal information to use and/or sell to make a profit

Potential Targets: Internet users and businesses

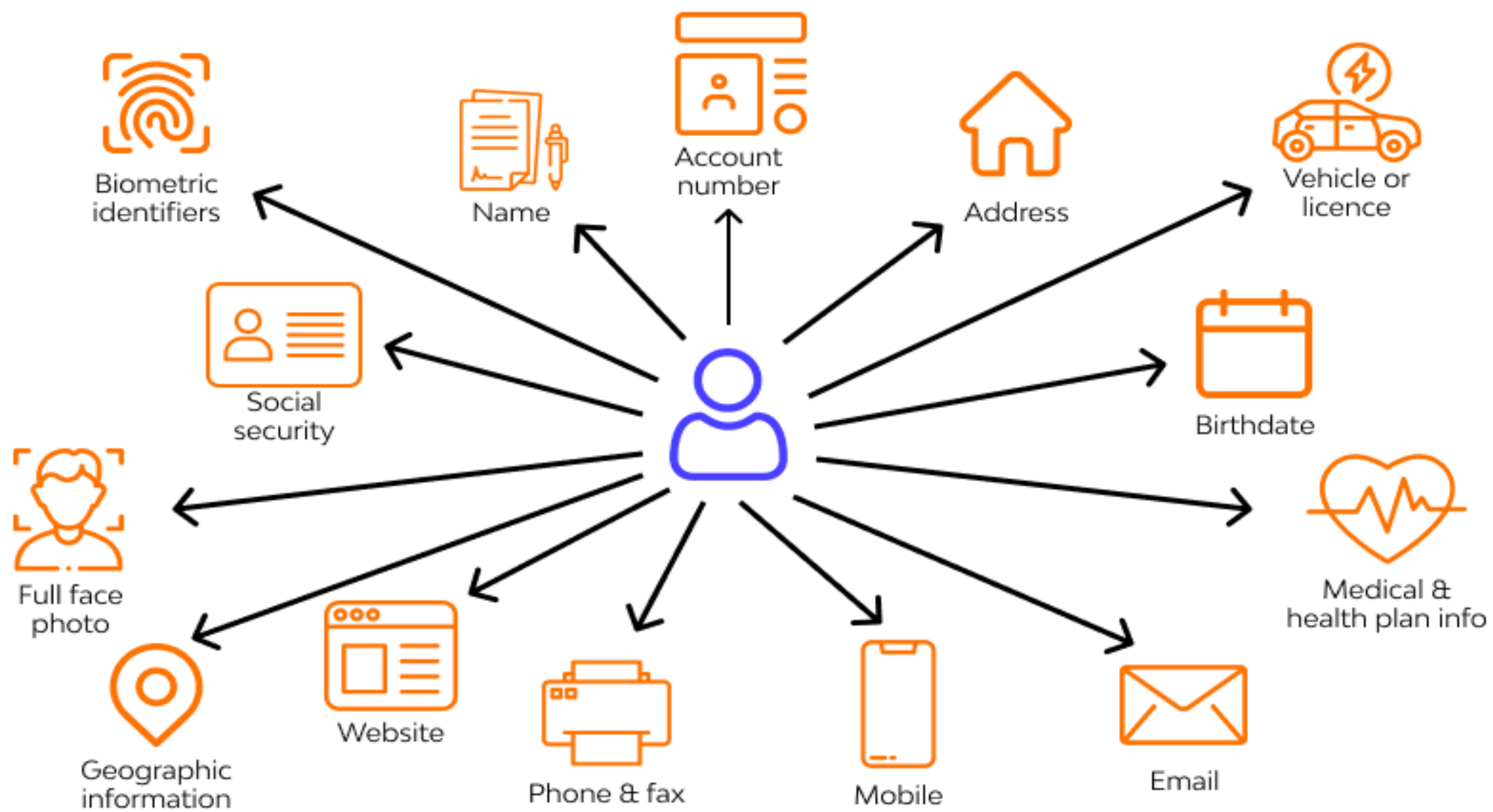
- countermeasure
- is an action, device, procedure, or technique that
- reduces a threat, vulnerability, or attack, eliminating or preventing it by minimizing the harm it can cause.
- It can also include discovering and reporting vulnerabilities
- so that corrective action can be taken.



Unit VI	Cyber Forensic, Hacking& its countermeasures	(08 Hours)
Personally Identifiable Information (PII), Cyber Stalking, Cybercrime, PII Confidentiality Safeguards, Information Protection Law: Indian Perspective. Hacking: Remote connectivity and VoIP hacking, Wireless Hacking, Mobile Hacking, countermeasures		

PII

- Any information used to distinguish or trace an individual's identity.
- Ex: name, social security number, date and place of birth, mother's maiden name, or biometric records;
- OR....
- any information that is linked or linkable to an individual.
- Ex: medical, educational, financial, and employment information.



Personally Identifiable Information (PII)

PII is defined as **information**:

- (i) that directly **identifies an individual** (e.g., name, address, social security number, telephone number, email address, etc.) or
- (ii) an **agency intends to identify specific individuals in conjunction with other data elements**, i.e., **indirect identification**. (Ex: gender, birth date, geographic indicator, and other descriptors).
- information **permitting the physical or online contacting of a specific individual** is also considered as personally identifiable information.
- This information can be maintained in **either paper, electronic or other media**.

- Department of Labor (DOL) reminds safeguarding sensitive information to take it seriously at all times.
- DOL internal policy specifies protection of PII and other sensitive data.
- Users must adhere to the rules of behavior defined in applicable Systems Security Plans, DOL and agency guidance.
- DOL contractors respect the confidentiality of such information.
- Contract employees also shall avoid office gossip and should not permit any unauthorized viewing of records contained in a DOL system of records.
- Only individuals who have a "need to know" in their official capacity shall have access to such systems of records.

With these responsibilities contractors should ensure that their employees:

- To have **information access at all times**.
- Obtain **DOL management's written approval** prior to taking any DOL sensitive information away from the office.
- The DOL manager's approval must **identify the business necessity** for removing such information from the DOL facility.
- When approval is **granted to take sensitive information** away from the office.
- the employee must **adhere to the security policies** described above.

Determine PII Confidentiality Impact Levels

- 1. **Identifiability:** Evaluate how easily PII can be used to identify specific individuals
- 2. **Quantity of PII:** Consider how many individuals are identified in the information (such as number of records).
- 3. **Data Field Sensitivity:** Evaluate the sensitivity of each individual PII data field. An individual's Social Security Number, medical or financial account information is generally considered more sensitive.
- 4. **Context of Use:** Context of Use is defined as the purpose, for which PII is collected, stored, used, processed, disclose, or disseminated.
- 5. **Obligation to Protect Confidentiality:** Many organizations are subject to laws, regulations, or other mandates governing the obligation to protect PII, such as the Privacy Act 1974, HIPAA of 1996, and GDPR etc.
- 6. **Access to and Location of PII:** Consider the nature of authorization access to PII, whether PII is being stored on or access from outside the direct control of the organization.

Security Breach: Confidentiality, Integrity, OR Availability (CIA)		
POTENTIAL IMPACTS	LOW	If the loss of confidentiality, integrity, or availability could be expected to have a limited adverse effect on organization operations, organization assets, or individuals. For example, the loss of CIA might i. Cause degradation in mission capability to an extent and duration that the organization is able to perform its primary functions, but the effectiveness of the functions is noticeably reduced ii. Result in minor damage to organizational assets iii. Result in minor financial loss iv. Result in minor harm to individuals
	MODERATE	If the loss of confidentiality, integrity, or availability could be expected to have a serious adverse effect on organization operations, organization assets, or individuals. For example, the loss of CIA might i. Cause a significant degradation in mission capability to an extent and duration that the organization is able to perform its primary functions, but the effectiveness of the functions is noticeably reduced ii. Result in a significant damage to organizational assets iii. Result in a significant financial loss iv. Result in a significant harm to individuals that does not involve loss of life or serious life threatening injuries.
	HIGH	If the loss of confidentiality, integrity, or availability could be expected to have a severe or catastrophic adverse effect on organization operations, organization assets, or individuals. For example, the loss of CIA might i. Cause severe degradation in mission capability to an extent and duration that the organization is able to perform its primary functions, but the effectiveness of the functions is noticeably reduced ii. Result in major damage to organizational assets iii. Result in major financial loss iv. Result in severe and catastrophic harm to individuals involving loss of life or serious life threatening injuries.

Safeguards for Confidentiality

- PII should be protected through a combination of measures, including **Operational Safeguards, privacy-specific safeguards, and Security Controls.**
- Organization should use a risk-based approach for protecting the confidentiality of PII.
- **1. Operational Safeguards**
 - a. **Create Policy and Procedure**
 - i. Access **rules** for PII within a system
 - ii. PII retention **schedules and procedures**
 - iii. **Privacy** in the system development life cycle process
 - iv. **Limitation** of collection, disclosure, sharing and use of PII
 - v. PII **incident response** and data breach **notification.**
 - vi. **Consequences for failure to follow privacy** rules of behavior.

- b. Awareness, Training, and Education
 - i. The definition of PII
 - ii. Applicable privacy laws, regulation, and policies
 - iii. Roles and responsibilities for using and protecting PII
 - iv. Appropriate disposal of PII
 - v. Sanctions for misuse of PII
 - vi. Recognition of a security or privacy incident involving PII
 - vii. Roles and responsibilities in responding to PII-related incidents and reporting

- **2. Privacy-Specific Safeguards**
- a. Minimize the **Use, Collection, and Retention of PII**
- Organization should consider the total amount of PII used, collected, and maintained, as well as the types and categories of PII as long as it is required for the current business.
- b. Conduct **Privacy Impact Assessments**
- PIA should address confidentiality risks at every stage of the system development life cycle (SDLC).
- c. **De-Identifying Information**
- **to describe records-** that PII removed, or masked so that remaining PII does not identify an individual.
- De-identified information can be re-identified by using code, algorithms that is assign to individual records.
- d. Anonymizing Information
- It involves the techniques to ensure the **data cannot be re-identified**.

- **3. Security Controls**
- *a. Access Control Enforcement*
- Organization can control access to PII through access control **policies and access enforcement mechanism.**
- *i. Separation of Duties*
- Organization can enforce separation of duties for **responsibilities** involving access to PII
- *ii. Least Privilege*
- Organizations can **enforce most restricted set of rights** needed by users to perform the specified task
- *iii. Remote Access*
- If remote access is permitted, the organization should ensure that the communications are encrypted
-

- *iv. User-Based Collaboration and Information Sharing*
- to assist users in determining whether access authorizations match access restrictions for PII
- *v. Access Control for Mobile Devices*
- Organizations can choose to prohibit or strictly limit access to PII from portable and mobile devices
- *b. Auditable Events*
- Organizations can **monitor events that affect** the confidentiality of PII, such as unauthorized access to PII
- *i. Audit, Review, Analysis and Reporting*
- Organizations can regularly review and analyze the information system's audit records for indications of inappropriate or unusual activity affecting PII.
- *c. Identification and Authentication*
- Users can be uniquely identified, and authenticated before accessing PII.

Cyber Crime

- Any **illegal activity** that involves a computer, another digital device or a computer network.
- It includes common cyber security threats like **social engineering, software vulnerability exploits and network attacks**. But it also includes **criminal acts** like **protests, harassment and extortion, money laundering, and more**.
- Cyber crime targets both **individuals and companies**.
- businesses for direct financial gain.
- large-scale scams
- to compromise their devices

Cybercrimes can generally be divided into two categories:

Crimes that target networks or devices

Viruses

Malware

DoS Attacks

Crimes using devices to participate in criminal activities

Phishing Emails

Cyberstalking

Identity Theft

Categories of Cybercrime

- There are three major categories that cybercrime falls into: individual, property and government.
- The types of methods used and difficulty levels vary depending on the category.

Property:

- This is similar to a real-life instance of a criminal illegally possessing an individual's bank or credit card details.
- The hacker steals a person's bank details to gain access to funds, make purchases online or run phishing scams to get people to give away their information.
- They could also use a malicious software to gain access to a web page with confidential information.

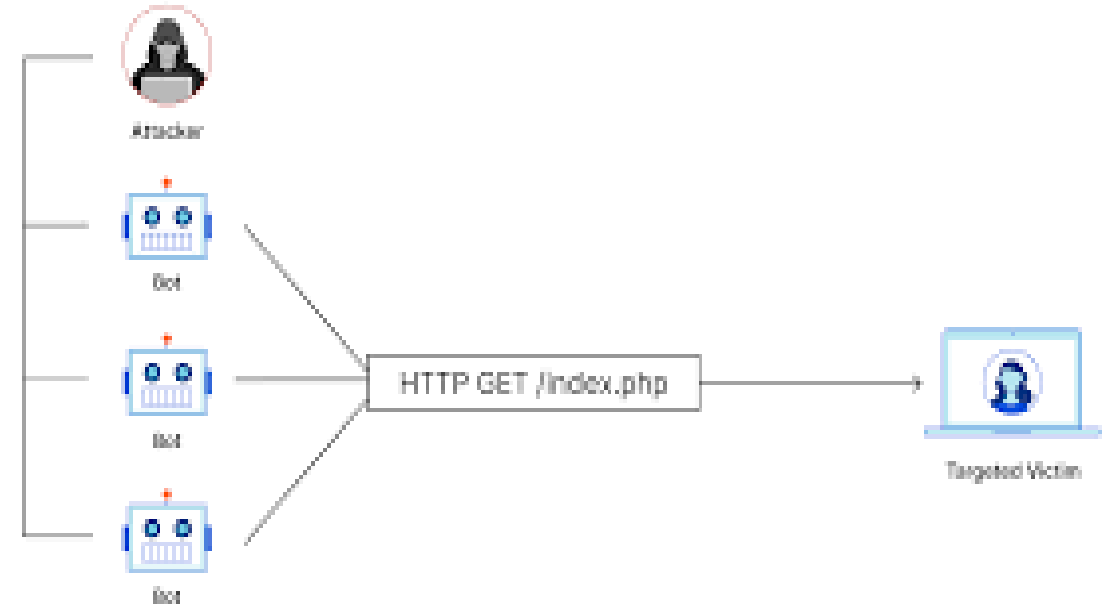
Individual:

- This category of cybercrime involves one individual **distributing malicious or illegal information online.**
- This can include cyber-stalking, distributing pornography and trafficking.

Government:

- This is the least common cybercrime, but is the most serious offense.
- A crime against the government is also known as **cyber terrorism**.
- Government cybercrime includes **hacking government websites, military websites**.
- These criminals are usually **terrorists or enemy governments of other nations**.

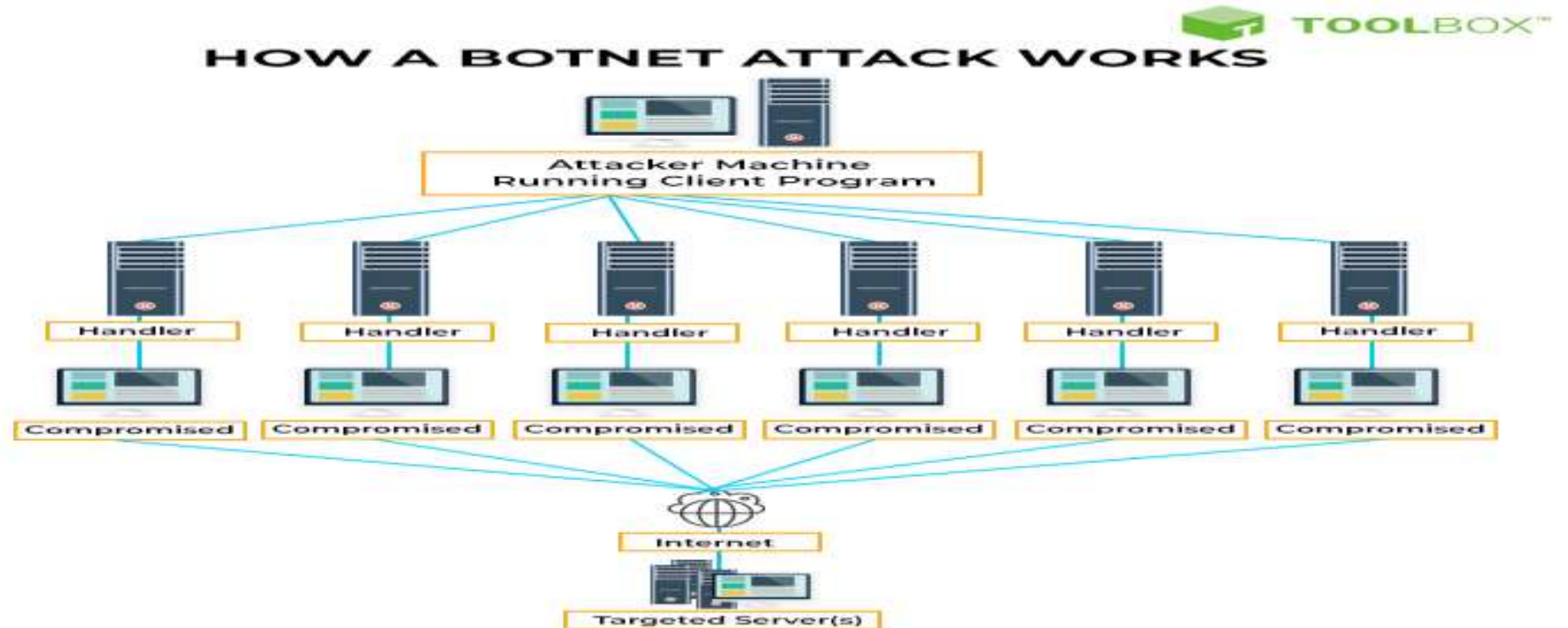
Types of Cybercrime



- **DDoS Attacks**
- DDoS attacks are used to make an **online service unavailable and take the network down** by overwhelming the site with traffic from a variety of sources.
- Large networks of infected devices known as **Botnets** are created by depositing malware on users' computers.
- The hacker then hacks into the system once the network is down.

- **Botnets**

- Botnets are **networks** from **compromised computers** that are controlled **externally by remote hackers**.
- The remote hackers then **send spam or attack other computers** through these botnets.
- Botnets can also be used to act as **malware and perform malicious tasks**.



- Identity Theft
- when a **criminal gains access to a user's personal information**
- to steal funds, access confidential information, or participate in tax or health insurance fraud.
- open a **phone/internet account** in your name, use your name **to plan a criminal activity** and **claim government benefits** in your name.
- They may do this **by finding out user's passwords** through hacking, retrieving personal information from social media, or sending phishing emails.



- **Cyberstalking**

- online harassment where the user is subjected by **online messages and emails**.
- cyberstalkers use social media, websites and search engines to **intimidate a user and instill fear**.
- Cyber-stalker knows their victim and makes the person feel afraid or concerned for their safety.



- **Social Engineering**

- Criminals making direct contact with you usually by phone or email.
- They want to gain your confidence and usually pose as a customer service agent so you'll give the necessary information needed.
- This is typically a password, the company you work for, or bank information.
- Cybercriminals will find out what they can about you on the internet and then attempt to add you as a friend on social accounts.
- Once they gain access to an account, they can sell your information or secure accounts in your name.

SOCIAL ENGINEERING



- **PUPs**

- PUPS or Potentially Unwanted Programs are **less threatening than other cybercrimes**, but are a type of malware.
- They uninstall necessary software in your system including **search engines and pre-downloaded apps**.
- They can include **spyware** or **adware**, so it's a good idea to install an **antivirus** software **to avoid the malicious download**.



- **Phishing**

- This type of attack involves **hackers sending malicious email attachments** or URLs to users to gain access to their accounts or computer.
- **Cybercriminals are becoming more established** and many of these [emails are not flagged as spam](#).
- Users are tricked into emails **claiming they need to change their password or update their billing information, giving criminals access.**



- **Prohibited/Illegal Content**

- criminals sharing and distributing inappropriate content that can be considered highly distressing and offensive.
- Offensive content can include, but is **not limited to, videos with intense violent and videos of criminal activity.**
- Illegal content includes materials advocating terrorism-related acts and child **exploitation material.**
- This type of content exists both on the **everyday internet** and on the dark web, an **anonymous network.**



- Online Scams

- ads or spam emails that include promises of rewards or offers of unrealistic amounts of money.
- Online scams include exciting offers that are “too good to be true” and when clicked on can cause malware to interfere and compromise information.



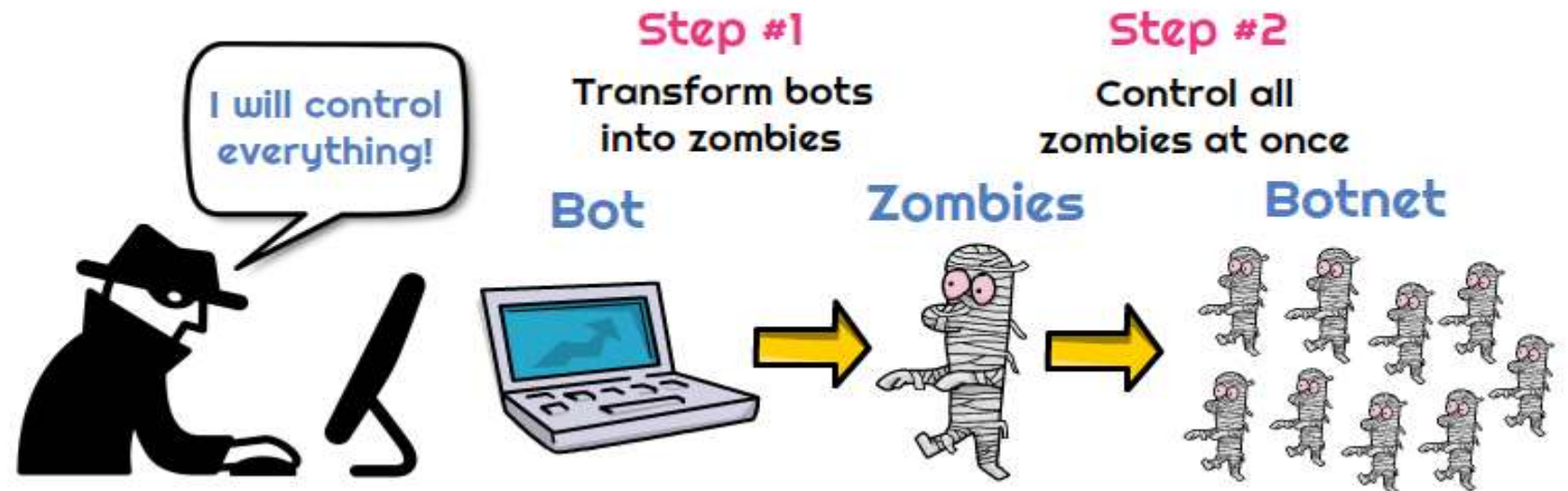
How to Fight Cybercrime

- 1 Become vigilant when browsing websites.
- 2 Flag and report suspicious emails.
- 3 Never click on unfamiliar links or ads.
- 4 Use a VPN whenever possible.
- 5 Ensure websites are safe before entering credentials.
- 6 Keep antivirus/application systems up to date.
- 7 Use strong passwords with 14+ characters.

zombie botnet

- When multiple computers get infected with the same malicious software.
- Malware consist of component that allows the attacker to control the infected computers remotely, typically via a command and control server

Bots, Zombies and Botnets



What is Cyberstalking?

- Cyberstalking is a type of cybercrime that uses the internet and technology to **harass or stalk a person**.
- it takes the form of **text messages, e-mails, social media posts, and other mediums**.
- Cyberstalking often starts with harmless interactions that **become systematic in an annoying or frightening manner**.
- At initial stage of cyberstalking to be amusing and harmless
- It not end even after the recipient has expressed their displeasure and asked for the interaction to stop.

- A cyber stalker might terrorize a victim by sending messages several times a day and from different accounts.
- Cyberstalking not involve direct communication.
- some victims not even realize that they are the victims of online stalking.
- The victims can be monitored through various methods and the information gathered can be later used for crimes such as identity theft.
- Some stalkers harassing the victims offline and even contacting their friends.

- characteristics of Cyberstalking behavior
- tracking locations,
- breaching [data privacy](#),
- monitoring online and real-world activities,
- obsessively tracking the victims' Social media by sending threatening private messages or faking photos.
- false accusations, spread malicious rumors, create fake social media profiles or blogs, or create and publish revenge porn.
- There might be a misperception that because it does not involve physical contact,
- Cyberstalking is not as severe as the physical form of stalking.

Signs of Cyberstalking

- someone sending you too many messages,
- a person sending you inappropriate messages,
- liking all of your old posts on social media, manipulating you into interacting with them online, or trolling you.
- Online impersonality, GPS tracking, threatening messages,
- cat fishing.

Cyberstalking Examples

- Posting offensive, suggestive, or rude comments online
- Joining the same groups and forums as the victim
- Releasing the victim's confidential information online
- Tracking all online movements of the victim
- Using technology for blackmailing or threatening the victim
- Excessively tagging the victim in irrelevant posts
- Engaging with all online posts made by the victim
- Creating fake profiles on social media to follow the victim
- Posting or distributing real or fake photos of the victim
- Making fake posts intended to shame the victim
- Repeatedly messaging the victim
- Hacking into the victim's online accounts
- Sending unwanted gifts or items to the victim
- Using [hacking tools](#) to get into the victim's laptop or smartphone camera and secretly record them
- Continuing harassment even after being asked to stop

Consequences of Cyberstalking

- affect to the victims both physically and mentally.
- Victims experience fear, anger, confusion, and insomnia along with other health issues.
- Cyberstalking affects the overall well-being of victims.
- anxiety, distress, depression, and suicidal ideation.
- You can contact a loved one,
- see a mental health professional,
- seek legal help, or
- even file a police complaint against the stalker

Difference between Cyberstalking and Cyberbullying

- The victim is harassed online through electronic communication devices, instant messaging, social networks, discussion groups, etc., for the purpose of **revenge, anger, or control**. A stalker may be a stranger or an known of the victim.
- Cyberbullying is mostly when a child, preteen, or teen is harassed, humiliated, tormented, threatened, embarrassed, or targeted through the internet, interactive and digital technologies, or electronic devices by another individual of the same age range.
- If adults are involved, it is considered **cyber-harassment or Cyberstalking**.

What are the main types of cyberstalking?

- **four main types:**
- vindictive,
- composed,
- intimate, and
- collective.
- **Vindictive** cyberstalking involves **threats**,
- **composed** cyberstalking involves **annoyance and harassment**,
- intimate involves **exes or people infatuated with the victim**,
- collective involves a **person being cyber stalked** by a group of individuals.

Types of Cyberstalking

Catfishing

- The creation of fake profiles or copying of existing ones on social media to approach victims.



Monitoring check-ins on social media

- Keeping an eye on the activities of a victim on social media to accurately gauge their behavior pattern.



Spying via Google Maps and Google Street View

- Using Street View to spy on a victim and find their location from posts or photos on social media.
- **Thai Villagers Arrest a Google Street View Driver, Thought He Was a Spy Photog**



Hijacking webcam

- Webcams can be hijacked by introducing malware-infected files into the victim's computer.



Webcam Hacking Explained



A hacker
targets
your device



They infect
your computer
with a virus



They gain
control of your
webcam

Installing stalkerware

- To a broad group of apps that someone else can install on your device to intercept texts and phone calls, access your location, log your web browsing activity and turn on your camera or microphone.
- The information gathered by such an app typically gets sent to a portal or companion app accessed by the person who installed the stalkerware.



Tracking location with geotags

- Digital pictures mostly have geotagged with the time and location of the picture if it is in the metadata format, which makes it easier for stalkers to access that information by using special apps.



GEOTAG

- 1 In this method consists of focusing a user around a location.
- 2 This includes country, state, zipcode and IP addresses.



GEOFENCE

- 1 In this method consists of drawing a fence around a location.
- 2 Once a user either enter or exist they can receive an alert on their phone for a promotion or coupon.

Legal Implications of Cyberstalking

Punishment for Cyberstalking



- India is at the top of the statistics when it comes to global sexual harassment.
- The online harassment faced by women also reflects the harassment in the physical world.
- 50 percent of women in the major cities of India have faced online abuse according to a survey by Feminism.
- Cyberstalking against men is also becoming quite common nowadays, making the ratio almost 50:50.

cyber-stalking laws in India:

- Any act of stalking is considered an offense committed. An act of stalking can be following, contacting, or attempting to contact a person repeatedly **despite clear indications of disinterest by them**, monitoring online activities of a person, or spying on a person that causes fear of violence, serious alarm, disruption of mental peace, or distress.
- The victim can file a case of **defamation against the offender**.
- Acts of stalking performed by someone who has been **authorized by the state for the purpose of preventing and detecting crime** are excused.

- Criminal intimidation made to any person with an injury to their reputation to cause alarm or to make them change their course of action on anything is considered to be a punishable crime.
- If someone is constantly harassed with derogatory verbal abuse because of gender issues, the perpetrator can be punished.
- Any criminal intimidation by anonymous communication or vengeful posting of videos, images, or images of rape victims is punishable with imprisonment.
- Non-consensual capturing of an image of a woman engaging in a private act and/or disseminating the said image is punishable by law.
- Online sexual harassment is punishable by law with up to three years of imprisonment and/or fine.

How to deal with Cyberstalking?

how you can deal with it:

- Send a clear indication in **writing form**.
- After it **do not engage** with the stalker at all.
- Go to the **police** if the harassment does not stop.
- Get your devices **checked for spyware** or **signs of compromised accounts**.
- If you think you are being tracked by someone through spyware, **use a family member's or friend's phone to get help**.
- Change **all passwords**.

- Use privacy settings to block the person off your social media, and report the abuse to the network.
- Filter abusive emails to a separate folder to avoid reading them.
- If you know the stalker's ISP, the part after the @ in their email address, contact abuse@domainname or postmaster@domainname.
- Google has a support system in place for such cases, <https://support.google.com/mail/contact/abuse>
- Make sure to have copies of any communication involved, police reports, and emails from the networks.
- Back up the evidence on an external drive.

How to avoid Cyberstalking?

**Increasing your privacy settings.
Enable strong privacy settings.**

- Make all posts **viewable to friends only**
- Do not enable permissions for social networks to post your contact details publicly.
- Try to have a **separate email address for social media** and other online activities.
- Share private information with friends over a private message rather than a **public post**.
- Avoid using your **real name** and use a gender-neutral screen name instead for your social media accounts.
- Leave **optional fields** in social media profiles blank.
- **Only accept friend requests** from those you know in person.
- **Disable geo-location settings** and GPS on your device.

How to Prevent Cyberstalking

- Make Security a Priority
- **Create strong passwords**
- **Be sure to log out every time.**
- **Keep track of your devices.**
- **Use caution on public wifi.**
- **Practice online safety habits.**
- Practice Good Digital Hygiene
- **Take advantage of security settings.**
- **Create generic screen names.**
- **Keep locations secure.**
- **Be careful with online dating sites.**
- **Perform a social media audit.**

How to Cope With Cyberstalking

- Set Up Roadblocks
- **Tell the person to stop.**
- **Block the person.**
- **Refuse to respond to any contact.**
- **Change email address and screen names**
- Increase Security
- **Change passwords.**
- **Consider suspending online accounts.**
- **Get your devices checked**
- Take Action
- **Save evidence of everything**
- **Notify your local police.**
- **Report them to the site or service they used.**

IT Act, 2000

- The Information Technology Act, 2000 was enacted by the **Indian Parliament in 2000.**
- It is the primary law in India for **matters related to cybercrime and e-commerce.**
- The act was enacted
 - to give legal sanction to electronic commerce and electronic transactions,
 - to enable e-governance, and also to prevent cybercrime.
- Under this law, for any crime involving **a computer or a network located in India**, foreign nationals can also be charged.
- The law prescribes **penalties for various cybercrimes** and fraud through digital/electronic format.
- It also gives **legal recognition to digital signatures.**

- The IT Act also amended certain provisions of the [Indian Penal Code \(IPC\)](#), the Banker's Book Evidence Act, 1891, the Indian Evidence Act, 1872 and the Reserve Bank of India Act, 1934
- to modify these laws to make them compliant with new digital technologies.
- In the wake of the recent Indo-China border clash,
- the Government of India banned various Chinese apps under the Information Technology Act.,
- ['TikTok, Other Chinese Apps Banned'](#).

IT Act – 2008 Amendments

- The IT Act, 2000 was amended in 2008.
- This amendment introduced the controversial Section 66A into the Act.

Section 66A

- Section 66A gave authorities the power to arrest anyone accused of posting content on social media that could be deemed 'offensive'.
- This amendment was passed in the Parliament without any debate.
- As per the said section, a person could be convicted if proved on the charges of sending any 'information that is grossly offensive or has menacing character'.
- It also made it an offence to send any information that the sender knows to be false, but for the purpose of annoyance, inconvenience, danger, obstruction, insult, injury, criminal intimidation, enmity, hatred or ill-will, through a computer or electronic device.
- The penalty prescribed for the above was up to three years' imprisonment with fine.

Arguments against Section 66A

- Experts stated that the terms 'offensive', 'menacing', 'annoyance', etc. were vague and ill-defined or not defined at all.
- Anything could be construed(interpreted) as offensive by anybody.
- There was a lot of scope for abuse of power using this provision to intimidate people working in the media.
- This also curbed the freedom of speech and expressed as a fundamental right in the Constitution.
- The section was used most notably to arrest persons who made any uncharitable remarks or criticisms against politicians.

Frequently Asked Questions Related to Information Technology Act 2000

Q1 What is the main provision of IT Act 2000?

- The original act addressed **electronic documents, e-signatures, and authentication of those records**. It also enacted **penalties for security breach offenses** including damaging computer systems or committing cyber terrorism.

Q2 What are the features of IT Act 2000?

- All **electronic contracts created through secure electronic channels** were legally valid.
- Legal **recognition for digital signatures**.
- Security measures for electronic records and **conjointly digital signatures** are in place.

Q3 How many sections are in the IT Act 2000?

- The original Act contained **94 sections**, divided into **13 chapters and 4 schedules**.

Hacking

- act of compromising digital devices and networks through unauthorized access to an account or computer system.
- Hacking is not always a malicious act, but it is most commonly associated with illegal activity and data theft by cyber criminals.

Key drivers of Hacking

- There **are four key drivers** that lead to bad actors hacking websites or systems:
 - (1) **financial gain through the theft of credit card** details or by defrauding financial services,
 - (2) **corporate espionage**(the activity of **collecting and reporting information**)
 - (3) to gain respect for their **hacking talents**, and
 - (4) **state-sponsored hacking** that aims **to steal business information** and national intelligence.
 - (5) there are **politically motivated hackers**—or **hacktivists**—who aim to raise public attention by leaking sensitive information.

Types Of Hackers



Grey Hat Hackers



Black Hat Hackers



White Hat Hackers

TYPES OF HACKERS:

White Hat vs. Black Hat & Every Shade in Between



Black Hat Hackers

- [Black hat hackers](#) are the "bad guys" of the hacking scene.
- They go out of their way to discover vulnerabilities in computer systems and software to exploit them for financial gain or for more malicious purposes,
- such as to gain reputation, carry out corporate espionage, or as part of a nation-state hacking campaign.
- Due to this serious damage on both computer users and the organizations occurs.
- They can steal sensitive personal information,
- compromise computer and
- financial systems, and alter or
- take down the functionality of websites and critical networks.

White Hat Hackers

- White hat hackers are the “good guys” who attempt to prevent the success of black hat hackers through
- [proactive hacking](#).
- It is the authorized entry into a company's network by professionals.
- Ethical Hacking:
- Use technical skills to break into systems to assess and test the level of [network security](#), known as [ethical hacking](#).
- This helps expose vulnerabilities in systems before black hat hackers can detect and exploit them.
- The techniques white hat hackers use are similar to or even identical to those of black hat hackers,
- but these individuals are hired by organizations to test and discover potential holes in their security defenses.

Grey Hat Hackers

- Grey hat hackers sit somewhere between the good and the bad guys.
- black hat hackers, they attempt to violate standards and principles but without intending to do harm or gain financially.
- Their actions are typically carried out for the common good.
- For example, they may exploit a vulnerability to raise awareness that it exists, but unlike white hat hackers, they do so publicly.
- This alerts malicious actors to the existence of the vulnerability.

What is VoIP hacking(Voice over Internet Protocol)

- VoIP hacking is a type of attack that a person uses **to infiltrate your business phone system.**
- listen in on calls,
- rack up(achieve) expensive bills, and
- steal sensitive information—your business *and* your customers.
- This might happens if one of **your employees accidentally gives information to a scammer.**
- These types of social engineering scams account for **97% of all malware attacks.**

What is Remote hacking

- A remote attack refers to a **malicious attack that targets one or more computers on a network.**
- Remote hackers look for **vulnerable points in a network's security**
- to remotely compromise systems,
- steal data, and
- cause many other kinds of problems.
- Some of the most types of remote attacks are:
 - 1. Domain Name System (DNS) Poisoning**
 - 2. Port Scanning**
 - 3. Password Spraying**
 - 4. Phishing**

Wireless Hacking

- A wireless network is a **set of two or more devices** connected with each other via **radio waves within a limited space range**.
- The devices in a wireless network have the freedom to be in motion, but be in connection with the network and share data with other devices in the network.
- One of the most crucial point that they are so spread is that their installation cost is **very cheap and fast** than the wire networks.

- Although wireless networks offer **great flexibility**, they have their **security problems**.
- A hacker can sniff the network packets without having to be in the same building where the network is located.
- As wireless networks communicate through **radio waves**, a hacker can easily **sniff the network from a nearby location**.
- Most attackers use **network sniffing to find the SSID** and hack a wireless network.
- Examples: Kismet, netstumbler, AirCrack-ng

Mobile Hacking

- Phone hacking involves any method where someone forces access into your phone or its communications.
- This can range from advanced security breaches to simply listening in on unsecured internet connections.
- It can also involve physical theft of your phone and forcibly hacking into it via methods like brute force.
- Phone hacking can happen to all kinds of phones, including Androids and iPhones.

How to Know If Someone is Hacking Your Phone

- **Your phone loses charge quickly.** Malware and **fraudulent apps** sometimes use **malicious code** that tends **to drain a lot of power**.
- **Your phone runs abnormally slowly.** A breached phone might be giving all its processing power over to the **hacker's shady applications**. This can cause your phone to slow to a crawl.
- Unexpected **freezing, crashes, and unexpected restarts** can sometimes be symptoms.
- **You notice strange activity on your other online accounts.** When a hacker gets into your phone, they will **try to steal access to your valuable accounts**.
- Check your **social media and email for password reset prompts**, unusual login locations **or new account signup verifications**.
- **You notice unfamiliar calls or texts in your logs.** Hackers may be tapping your phone with an **SMS Trojan**. Alternatively, they could be impersonating you to steal personal info from **your loved ones**.

Countermeasures/Prevention from Getting Hacked

- **Software Update**
- **Use Unique Passwords for Different Accounts**
- **HTTPS Encryption**
- **Avoid Clicking on Ads or Strange Links**
- **Change the Default Username and Password on Your Router and Smart Devices**

Protect Yourself Against Hacking

- **Install Antivirus Software**
- **Use a VPN**
- **Do Not Login as an Admin by Default**
- **Use a Password Manager**
- **Use Two-factor Authentication**
- **Brush Up on Anti-phishing Techniques**
- **Don't Let Hackers Drive You Out of Business**



Question Bank

1. Explain the classification of cybercrime in detail.
2. Explain the legal perspectives-indian perspective to tackle cybercrime.
3. What is social engineering? How to prevent from social engineering attacks?
4. Write a short note on Indian IT act.
5. Write a short note on challenges to indian law and cybercrime scenario in India.
6. Explain cyber stalking along with its types.
7. What are the common reasons for cyber crime?
8. Explain categories of cyber crimes.
9. What is password cracking and key loggers

Assignment No.6

1. Identify and explore the different types of Cyber stalker attacks.
2. Illustrate life cycle of cyber forensics?
3. List VoIP hacking types and explore any 3? What are the counter measures for it.
4. Who are cyber criminals? What are types of cyber crimes.
5. What is Botnet? How to protect from botnet?
6. Explain the terms:
 - i) Virus
 - ii) Phishing
 - iii) Spoofing
 - iv) Phone phishing
 - v) Internet pharming
 - vi) Cyber Forensic

Life Cycle of Cyber/Digital Forensics

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Identification

- Identify the purpose of investigation
- Identify the resources required

Preservation

- Data is isolate, secure and preserve

Analysis

- Identify tool and techniques to use
- Process data
- Interpret analysis results

Documentation

- Documentation of the crime scene along with photographing, sketching, and crime-scene mapping

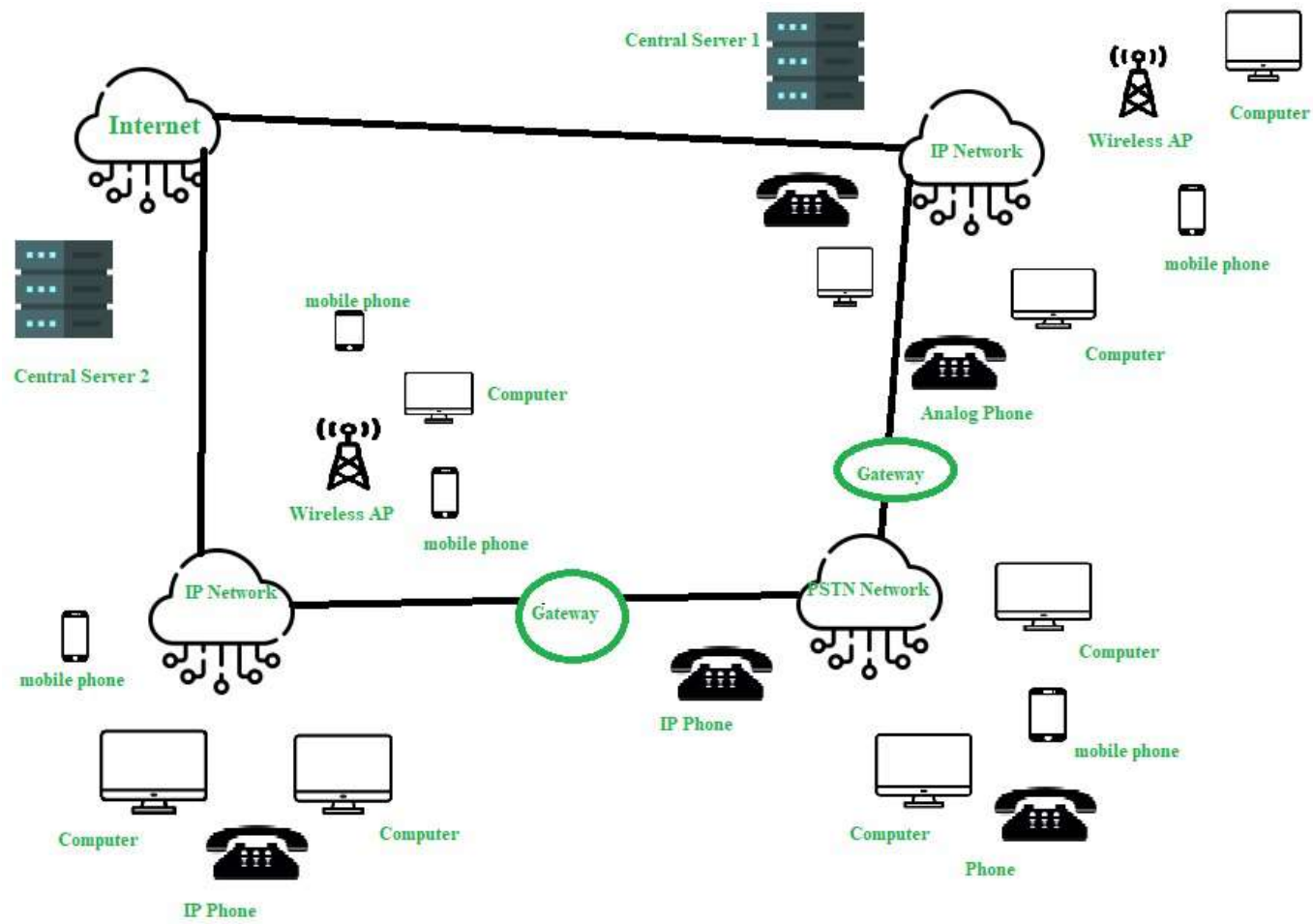
Presentation

- Process of summarization and explanation of conclusions is done with the help to gather facts.

List VoIP hacking types and explore any 3?

What are the counter measures for it

- Voice over IP or Voice over Internet Protocol (VoIP) is a collection of different technologies and practices that allows the delivery of voice communication, images, audio, video, through packet data networks over the internet protocol. This makes it very cost-efficient, flexible, and various other advantages as compared to the older telephonic systems.



- **VoIP Hacking:**
- Voice-over IP hacking is a type of attack carried out by the malicious user for the purpose of infiltrating the phone system or unauthorized access to the phone system in order to steal the data. This lets the malicious user listen to all the conversations and calls, steal critical information, to make calls and international calls to frame up huge bills. These attacks usually happen when an insider unknowingly gives out information or conspires with the malicious actors. The severity of these attacks could range from being totally useless to even dangerous enough to make the organization go bankrupt.

- **Types of VoIP Hacking:**
- **Unauthorized use:** Unauthorized use attack is when the malicious users make use of the organizations' phone network to make calls to other people or organizations pretending to be someone from the organization. The unauthorized malicious users use auto-dialing and robocalling software with the organization's phone network system. When a recipient picks up the call, a prerecorded message plays which might ask them to do some ask such as calling the malicious users, entering their information or entering their bank or credit card details, etc. These can be countered by periodically monitoring the history, call logs, etc.
- **Toll fraud:** When these malicious actors make international calls to other people and organizations. As the charge of these calls can be fairly expensive and the bill that will be charged from the organization's account can often be damaging. This is called toll fraud.
- **Spoofing of Caller ID:** In this type of attack, the malicious actors use forged caller IDs and control them in coordination with other attacks.
- **Eavesdropping:** Eavesdropping is when an attacker listens to the business calls, conversations, voicemails, without the user's knowledge. It usually only happens when the data is unencrypted or shared over an unencrypted and unsecured channel.

- **Call Tampering:** Incall tempering malicious actor tempers a phone call. For example, the attacker can inject some noise packets to make the quality of the call bad, or the attacker could hold back the packet delivery in order to create disturbance in communication.
- **Buffer Overflow Attacks:** A buffer is a temporary area for data storage. When more data (than was originally allocated to be stored) gets placed by a program or system process, the extra data overflow. It causes some of that data to leak out into other buffers, which can corrupt or overwrite whatever data they were holding. In a buffer-overflow attack, the extra data sometimes holds specific instructions for actions intended by a hacker or malicious user; for example, the data could trigger a response that damages files, change data, or unveils private information.
- **DoS Attacks:** A DDoS attack, the attacker tries to make a particular service unavailable by directing continuous and huge traffic from multiple end systems.
- **Viruses and malware:** A virus is a fragment of code embedded in a legitimate program. Viruses are self-replicating and are designed to infect other programs. Malware is software that gets into the system without user consent with an intention to steal private and confidential data of the user that including bank details and passwords.
- **Man-in-the-middle attacks:** Man In The Middle Attack implies an active attack where the attacker/Hacker creates a connection between the victims and sends messages between them or may capture all the data packets from the victims.

- **Countermeasures:**
- Choose a trusted VoIP provider with a good track record.
- Admin access should be implemented in a controlled and careful manner.
- VPNs should be used in case of remote access.
- Detailed network tests should be done periodically.
- VPNs should be used and their endpoint filtering should be enabled.
- Regularly checking history, access, and call logs.
- Passwords used should be strong.
- 2FA should be enabled.
- VoIP firmware and OS should always be up to date.
- Organizations should have more security professionals.

Who are **Cybercriminals**?

- They are individuals or teams of people who use technology to commit malicious activities on digital systems or networks with the intention of stealing sensitive company information or personal data, and generating profit.
- Cybercriminals are known to access the cybercriminal underground markets found in the deep web to trade malicious goods and services, such as hacking tools and stolen data. Cybercriminal underground markets are known to specialize in certain products or services.
- Laws related to cybercrime continue to evolve across various countries worldwide. Law enforcement agencies are also continually challenged when it comes to finding, arresting, charging, and proving cybercrimes.
- cyber criminals

What is Botnet? How to protect from botnet?

- <https://www.pingidentity.com/en/resources/cybersecurity-fundamentals/threats/botnet-attack.html#:~:text=Botnets%20are%20designed%20to%20exploit,s oon%20as%20they%20become%20available.>

Explain the term:

- 1. virus
- is a type of malicious software, or malware, that spreads between computers and causes damage to data and software. Computer viruses aim to disrupt systems, cause major operational issues, and result in data loss and leakage.

- 2. Phishing
- are the practice of sending fraudulent communications that appear to come from a reputable source. It is usually done through email. The goal is to steal sensitive data like credit card and login information, or to install malware on the victim's machine.

- 3. Spoofing:
- Spoofing, as it pertains to cybersecurity, is when someone or something pretends to be something else in an attempt to gain our confidence, get access to our systems, steal data, steal money, or spread malware. Spoofing attacks come in many forms, including: Email spoofing.

4. Phone Phishing:

is a type of social engineering and cybersecurity attack where the attacker impersonates someone else via email or other electronic communication methods, including social networks and Short Message Service (SMS) text messages, to reveal sensitive information.

- 5. Pharming:
- is online fraud that involves the use of malicious code to direct victims to spoofed websites in an attempt to steal their credentials and data. Pharming is a two-step process that begins with an attacker installing malicious code on a victim's computer or server.

6. Computer forensics:

also called digital or cyber forensics, is a field of technology that uses investigation techniques to help identify, collect, and store evidence from an electronic device.

What is password cracking and key loggers

- <https://courseware.cutm.ac.in/wp-content/uploads/2020/06/Cracking-and-Keyloggers.pdf>